

■ Results & Performance Analysis

Sensor accuracy, actuator response, and system validation results

System Test Results Summary

The Scuba Diver Health Monitoring System was tested in a controlled underwater environment. All sensors, actuators, and communication modules were evaluated for accuracy, response time, and reliability. The following results were obtained:

Sensor Accuracy Results

Parameter	Sensor Used	Measured Accuracy	Acceptable Range	Result
Heart Rate	MAX30102	±2–3 BPM	±5 BPM	■ Accurate
SpO2 (Oxygen)	MAX30102 PPG	±2%	±3%	■ Accurate
Body Temperature	DS18B20	±0.2°C	±0.5°C	■■ Highly Accurate
Li-Fi Data Loss	LED + Photodiode	2%	<5%	■ Stable

Actuator Response Time Results

Component	Function	Response Time	Trigger Condition	Result
Buzzer	Audio alert	0.5 sec	Abnormal vitals detected	■ Immediate warning
Relay Module	Controls air motor	1.2 sec	ESP32 detects critical values	■ No delay
Air Motor	Emergency balloon	1.8 sec	Low SpO2 / relay trigger	■■ Activated instantly
LED Alert	Visual warning on LCD	Instant	Any threshold breach	■ Functional
Panic Switch	Manual SOS	<0.3 sec	Diver presses button	■ Immediate SOS

Li-Fi Communication Performance

- Average communication latency: <100 ms (real-time monitoring confirmed)
- Packet loss: Near-zero under clear water conditions
- Working range: 2–3 meters line-of-sight underwater
- LED modulation: High optical output stability during prolonged operation
- Signal distortion: Minimal under clear water; increases in turbid water
- Compared to RF: Li-Fi far more reliable underwater; RF attenuates rapidly in seawater

Sensor Graphs on ThingSpeak Dashboard

Heartbeat Rate Graph (Fig 4.1)

Normal range 72–110 BPM. Clean PPG waveforms with distinct systolic/diastolic peaks. Error margin ±2 BPM.

Body Temperature Graph (Fig 4)	Smooth temperature transitions at ~36°C. No noise from water movement. Detected hypothermia
SpO2 / Pulse Oximeter Graph (Fig 5)	Clear systolic peaks, minimal distortion. SpO2 accuracy $\pm 2\%$, heart rate ± 3 BPM. Instantly detected

Advantages Demonstrated

- Reliable underwater communication using light waves (Li-Fi) vs. ineffective RF
- Real-time continuous monitoring of all three vital signs simultaneously
- Quick emergency response — automatic balloon inflation in under 2 seconds
- Dual alert system: surface buzzer + LCD display ensures team is notified
- Low electromagnetic interference — safe for sensitive underwater environments
- Cloud logging via ThingSpeak allows remote monitoring and post-dive analysis